

Wellmere

Season Lead · Wellmere Seed Bank

9 DAYS · 40 STOPS · GRADE 6 · SEEDBANK_MS

THE OPENING CARD

Two hundred kilometres east there is a wheat disease the breeding programme has exactly one gene against. The wind that would carry it here comes in over the causeway. You are Season Lead at Wellmere. Forty-one thousand seed samples are held here at minus eighteen, and the ground outside is laid out in rings, spaced by how far pollen can travel. Sowing starts in three weeks, and what goes into the ground is settled this fortnight. The trial has to be laid out. The oldest samples have to be grown out again, before fewer of them will come up. And a new wheat has to be either released to the merchant or held back. Anything not sown in three weeks waits a full year.

What is actually in the vault

BEFORE THE DAY — GREET

Six people, one season, and the crossing block is Thursday

Everybody here gets one growing season a year, and they all want the same glasshouse space. Get round and put a name to as many of them as you can before the plan is argued over. It is a shorter argument between people who have met.

PLAN CARD 73W

Monday in the first week of March, and the sowing plan has to be signed by Friday. Wellmere holds forty-one thousand seed batches at minus 18. Yusuf Mbaye says sixty of them must be grown out this season, and there is ground for about forty. Chiara Volpe wants that ground for her trial, and Nasrin Qureshi wants it for the collection. Today you find out what a seed does while it is stored.

BRIEFING 7W

41000 accessions, and 60 of them overdue.

WHAT YOU DECIDE 15W

Find out what Wellmere is holding and what it is doing while it sits there.

1.1 Why the vault is cold

CONSERVATION & VIABILITY · SEED VAULT · A ROOM · CHOICE

SCENE 34W

Mbaye has two lots of the same seed on the bench. One spent four years dried, at minus eighteen degrees. The other sat in a warm shed, and only three in ten still grow.

GUIDE 59W

Four accounts of why cold and dry works. Ask of each whether it treats the seed as alive or as dead. A dried seed is still slowly changing inside, so damage builds up over years. Two of these options have something being blocked or killed instead. And a clean sealed packet still lasts far longer when it is cold.

ASK 10W

Why does cold, dry storage keep a seed alive longer?

BEHIND THE BUTTON 169W

Why the wrong answers are worth reading. Each one is written to look right to somebody who made a particular mistake. Working out which mistake each one belongs to is most of the learning in this format — more than getting the right one, which you can sometimes do by feel.

Why sounding careful is not the same as being right. A long answer with a lot of detail in it feels safer, and it is easy to write one that is still wrong. The check that builds these questions measures option length across the whole game so the longest one is not the answer more often than chance. Reading is the only way through.

What the verdict adds. After you answer, the verdict says why each wrong option is wrong, in its own words. That is written for the reader who picked it. Read it even when you were right, because being right for a shaky reason is the thing that catches up with you two stops later.

TAKES AS READ

some seeds can survive being dried out and frozen

VERDICT 73W

A dried seed is still alive, and being alive means chemistry going on inside it. That chemistry slowly damages the parts the seed will need when it wakes up. Taking the water out slows those changes down. Cold slows them further, so the two work together. It is not a fuel tank running out. It is damage adding up, and everything in the vault is there to make it add up more slowly.

TAKEAWAY 14W

Keeping something alive for decades is a matter of slowing down what damages it.

1.2 Two parents, and what you are asking of them

HEREDITY & CROSSING · CROSSING HALL · A PERSON · CHOICE

SCENE 34W

Volpe has two parents pinned above the bench. One is a modern wheat that yields well. The other is an old local kind that came through the 2018 drought. Bengtsson asks what she wants.

GUIDE 57W

Four accounts of what a cross is for. Ask of each whether it could happen when each parent passes on half its genes. The offspring get a mix, not an average. Neither parent already has the combination being looked for. And many of the offspring will be worse than both, because the bad genes get shuffled too.

ASK 11W

What is the programme actually asking for by crossing these two?

BEHIND THE BUTTON 169W

Why the wrong answers are worth reading. Each one is written to look right to somebody who made a particular mistake. Working out which mistake each one belongs to is most of the learning in this format — more than getting the right one, which you can sometimes do by feel.

Why sounding careful is not the same as being right. A long answer with a lot of detail in it feels safer, and it is easy to write one that is still wrong. The check that builds these questions measures option length across the whole game so the longest one is not the answer more often than chance. Reading is the only way through.

What the verdict adds. After you answer, the verdict says why each wrong option is wrong, in its own words. That is written for the reader who picked it. Read it even when you were right, because being right for a shaky reason is the thing that catches up with you two stops later.

TAKES AS READ

young plants get half their genes from each parent

VERDICT 74W

Each parent hands on half its genes, so the first generation gets one set from each. The generation after that shuffles those sets into thousands of combinations. What the breeder wants is the yield genes of the modern parent together with the drought genes of the old one. Neither parent has that. Most of the offspring will be worse than either. Breeding is growing enough of them to find the rare ones worth keeping.

TAKEAWAY 19W

A cross is a request for a combination, and it takes generations to find out whether it was granted.

1.3 Why the same line is sown four times

FIELD TRIALS & SELECTION · FIELD LABORATORY · A ROOM · CHOICE

SCENE 31W

Meera Quiroga has the plot map on the wall. Two hundred and forty plots, each line sown four times across four blocks, in scrambled order. The north end is visibly wetter.

GUIDE 52W

Four things four plots might buy. Ask of each whether it is about the comparison or about safety. A field is not the same everywhere: drainage, depth and shelter all vary. So one plot's yield is the seed and the ground together. Spreading a line across the field is what separates them.

ASK 10W

What does sowing each line in four scattered plots buy?

BEHIND THE BUTTON 169W

Why the wrong answers are worth reading. Each one is written to look right to somebody who made a particular mistake. Working out which mistake each one belongs to is most of the learning in this format — more than getting the right one, which you can sometimes do by feel.

Why sounding careful is not the same as being right. A long answer with a lot of detail in it feels safer, and it is easy to write one that is still wrong. The check that builds these questions measures option length across the whole game so the longest one is not the answer more often than chance. Reading is the only way through.

What the verdict adds. After you answer, the verdict says why each wrong option is wrong, in its own words. That is written for the reader who picked it. Read it even when you were right, because being right for a shaky reason is the thing that catches up with you two stops later.

TAKES AS READ

the same crop grows differently in different parts of a field

VERDICT 76W

Ground varies. Drainage, depth, shelter, how hard it was driven over. Sow a line once and its yield is the seed and that patch mixed together, with no way to tell them apart. Sow it four times in different blocks and the patches differ while the seed does not. Scrambling the order stops anybody's habits handing good ground to a favourite. How much the four disagree is itself a measure of how much the ground matters.

TAKEAWAY 19W

Repeating a thing in different places is how you find out which part of the answer was the place.

A stored seed is alive and spending itself, and everything in the vault is aimed at slowing that down.

What the generation tells you

BEFORE THE DAY — TRIAL

Walk the station before the sowing plan is signed

The station is built in rings. The vault sits in the middle, then the laboratory, then the glasshouses. Walk the inner rings once, because a plan drawn from a map does not tell you which way the wind crosses the headland.

PLAN CARD 71W

Tuesday. Four hundred plants from Volpe's cross are in the warm bay. This is the generation where the parents' hidden versions come apart and show themselves. Hana Nakagawa has run a marker test on the same plants and has a disagreement to explain. Eleven of them carry the resistance gene and do not look it. Today you count what the generation is telling you, and work out what the mismatch means.

BRIEFING 11W

400 plants in the warm bay, and a ratio to count.

WHAT YOU DECIDE 14W

Read the second generation of last year's cross, and check it against a marker.

2.1 Counting a generation

HEREDITY & CROSSING · CROSSING HALL · A ROOM · BALLPARK

SCENE 29W

Four hundred plants from the second generation. Three hundred and six shrug the disease off, and ninety-four do not. The breeder asks what that number was supposed to be.

GUIDE 55W

Work out what number was expected before looking at what came. Each parent carries one resistant copy and one that is not, so three plants in four should shrug the disease off. That is a share of the four hundred. One tile is the count that actually came, which is what you are comparing against.

ASK 12W

How many of the four hundred plants should shrug the disease off?

BEHIND THE BUTTON 118W

Where three in four comes from. Each parent gives one of its two copies to each seed. There are four ways the copies can pair up, and three of those four contain a resistant copy. One resistant copy is enough, so three in four look resistant.

Why the whole generation is the evidence. Any single plant is resistant or not, and tells you nothing. Four hundred of them make a proportion, and a proportion is what a breeding rule predicts.

Why close is the right answer. Nobody expects exactly three hundred. Coin tosses do not come out exactly half, and four hundred plants is enough for the count to land near the expected number rather than on it.

TAKES AS READ

a plant carries two copies of each gene and passes on one of them

VERDICT 142W

Each parent has two copies of the gene and gives one to each seed. There are four ways those copies can pair up, and three of the four include a resistant copy. One copy is enough to shrug the disease off, so three plants in four should manage it. Three quarters of four hundred is $400 \times 0.75 = 300$. The tray gave 306, which is as close as anybody should expect — a proportion lands near its expected number, not exactly on it, in the same way a hundred coin tosses rarely give exactly fifty heads. That is what makes the count evidence. If two genes were doing the work, and either one was enough, the expected number would be 375 instead, and 306 is nowhere near it. No single plant in the tray could have told you any of this.

TAKEAWAY 13W

The proportions across a whole generation carry information that no single plant does.

2.2 What the screen may not find

MARKERS & GENE EXPRESSION · MOLECULAR LABORATORY · A PERSON · CHOICE

SCENE 34W

Nakagawa is screening fifty batches for a second resistance gene. Twenty-two of them are on the drifted list, and their test controls are clean. She wants to know what a clean result still means.

GUIDE 55W

Four readings of what drift does to a test. Ask of each whether it changes clean results, positive results, or both. Chance can remove a gene that was rare, so a clean result now has two stories and the test cannot tell them apart. Chance cannot invent the gene, so a positive still means something.

ASK 8W

What does the drift mean for the screen?

BEHIND THE BUTTON 169W

Why the wrong answers are worth reading. Each one is written to look right to somebody who made a particular mistake. Working out which mistake each one belongs to is most of the learning in this format — more than getting the right one, which you can sometimes do by feel.

Why sounding careful is not the same as being right. A long answer with a lot of detail in it feels safer, and it is easy to write one that is still wrong. The check that builds these questions measures option length across the whole game so the longest one is not the answer more often than chance. Reading is the only way through.

What the verdict adds. After you answer, the verdict says why each wrong option is wrong, in its own words. That is written for the reader who picked it. Read it even when you were right, because being right for a shaky reason is the thing that catches up with you two stops later.

TAKES AS READ

a gene that is rare in a population may be missing from a small sample of it

VERDICT 79W

Drift can remove a gene that was rare to begin with. So a clean result from one of those batches has two explanations. The gene was never there, or it was there and got lost in a grow-out. The test cannot tell those apart, because it is looking at the seed in front of it. A positive is still worth having, because drift removes genes and never invents them. The fix is to test material that missed the bottleneck.

TAKEAWAY 22W

A screen finds what is in front of it, and what is in front of it is what survived the last grow-out.

2.3 Where a tonne of grain comes from

AGRONOMY & PLANT ENERGETICS · DRYING & PROCESSING HALL · A ROOM · BALLPARK

SCENE 32W

The agronomist has eleven tonnes of grain from one field on the weighbridge slip. A trainee has just said the crop got it out of the soil. He would like that settled.

GUIDE 59W

Work out how much of the eleven tonnes is the stuff the plant built out of air. About nine tenths of dry grain is sugar and starch, and all of that came in through the leaves as carbon dioxide. One tile is the share that came out of the soil, which is the small part and not the answer.

ASK 15W

How much of the eleven tonnes came in through the leaves, out of the air?

BEHIND THE BUTTON 113W

What grain is made of. Once it is dry, about nine parts in ten are carbohydrate — sugars and starch, built out of carbon dioxide from the air and water, using light. That is where almost all the weight comes from.

What the soil gives. Minerals, and only a few per cent of the weight. They matter enormously, because a crop short of nitrogen cannot build much of anything, and they are not what the harvest is made of.

Where the water goes. Most of the water a crop takes up leaves again through its leaves as vapour. It is not in the weighbridge figure at all, because the grain is weighed dry.

TAKES AS READ

plants build sugars out of carbon dioxide and water, using light

VERDICT 155W

Dry grain is about nine parts in ten carbohydrate — sugar and starch. The plant builds all of it out of carbon dioxide taken in through its leaves, with water and light. So the part of the harvest that came out of the air is $11 \times 0.9 \approx 9.9$ tonnes, near enough ten. The soil's contribution is the other part: minerals, a few per cent of the weight, a few hundred kilograms across the whole field. That small share is not unimportant — a crop short of nitrogen cannot build much of anything — but it is not what the harvest is made of. The water is not in the figure at all. Most of what the crop took up left again through its leaves as vapour, and the grain on the slip has been weighed dry. So the trainee has it backwards: the field gave the minerals, and the air gave the weight.

TAKEAWAY 16W

A harvest weighs what it weighs because of something that was not there when it started.

2.4 Six per cent, while the lorries arrive

CONSERVATION & VIABILITY · SEED VAULT · A ROOM · HOLD

SCENE 35W

Seed has to be dried to six per cent before it goes into the vault. Wet lots keep arriving off the field. The dehumidifier is the only thing that takes water out of the air.

GUIDE 54W

Hold the room at six per cent, inside the band on the dial. The band gets narrower later in the week, because seed that is nearly dry is spoiled by less. Each wet lot keeps adding water for as long as it stands there. Set the dehumidifier to answer the load, not the reading.

ASK 14W

Hold the drying room at six per cent while the week's seed comes in.

BEHIND THE BUTTON 66W

Why a lot is not a single push. A lorry does not add water once and stop. The wet seed keeps giving water to the air while it sits in the room.

Why dry matters. A seed is alive. Cold and dry both slow it down, so it stays alive for longer. The freezer holds itself. The drying is held by whoever is on the floor.

TAKES AS READ

seed is dried before it goes into store

VERDICT 131W

Every lot that arrives is a load that keeps working. Wet seed gives up water to the air for as long as it stands in the room, so the reading climbs and keeps climbing. One nudge on the dehumidifier does nothing that lasts. Set it where it matches what is coming in, and leave it there. Later in the week the band narrows. Early on, the seed is wet and a tenth of a point either way makes no difference. Near six per cent it is the difference between seed that keeps for decades and seed that keeps for years. That is the whole job of a vault. Cold and dry each slow a seed down, and they multiply together. The freezer holds itself. The dryness is held by a person.

TAKEAWAY 11W

A load that keeps arriving is not answered by one correction.

END OF THE DAY 16W

A ratio is evidence about what the parents carried, not a decoration on a data sheet.

The season you do not get back

BEFORE THE DAY — FOLLOW

Quiroga walks the trial rows once

The trial is four blocks of twenty rows, and the map only helps if you know where each block starts. Quiroga, the field trials manager, walks it once, at scoring pace. Stay with her, because she stops at the row ends and that is where the map and the ground disagree.

PLAN CARD 70W

Thursday, and the sowing plan is due tomorrow. Volpe's line WM-712 has been through four seasons and yields eleven per cent over the standard in dry years. Releasing it needs the whole west ground — which is the ground Qureshi's sixty overdue batches would have used. The forecast for summer is dry. Today you work out what a season's delay actually costs, and what the trial data does not support.

BRIEFING 13W

WM-712 is ready. The ground it needs is the ground the vault wants.

WHAT YOU DECIDE 11W

Weigh a release against a delay, with a dry summer forecast.

3.1 An extra generation in the bay

HEREDITY & CROSSING · CROSSING HALL · A ROOM · SEQUENCE

SCENE 28W

Bengtsson can run one more generation over the winter in the warm bay. A generation there takes three months instead of twelve. There is room for one population.

GUIDE 47W

Three jobs want the bay, and the rail is not a clock. Put them in order by what the winter can never give back: the job that only works indoors goes first, and the job that the field will do just as well in spring goes last.

ASK 16W

Order the three jobs by how much the winter in the bay is worth to each.

BEHIND THE BUTTON 119W

What the bay is good for. Getting a generation finished out of season, so a cross moves one step closer while the fields are empty. Three months instead of twelve, four times a year instead of once.

What the bay is bad for. Judging anything. Plants in a warm bay grow in light, water and temperature nothing like a field, so a yield measured there is a fact about the bay. Any judgement has to wait for ground outside.

Why multiplying seed goes last. It works perfectly well in the bay and it works perfectly well in a field in spring, so spending the winter on it spends a scarce thing on a job that has an easy alternative.

TAKES AS READ

a glasshouse can finish a generation out of season

VERDICT 158W

The bay does one thing nothing else can do. It finishes a generation out of season, in three months instead of twelve. So the job that belongs there is the one where a generation is the whole point. That is the backcross. It needs one more round before it can be tested outside, and every winter it is not run adds a year to the programme. The overdue batch comes next. It does need regrowing, and it needs enough plants to keep what is in it, which the bay can just manage. Multiplying seed for the merchant goes last. Not because it does not matter — it works just as well in a field in spring. Putting it in the bay spends the scarce thing on the job with an easy alternative. And nothing in there can be judged. Yield in a warm bay is a fact about the bay, so any measuring still waits for ground outside.

TAKEAWAY 12W

A generation indoors is for moving material along, not for judging it.

3.2 What the trial can actually claim

FIELD TRIALS & SELECTION · FIELD LABORATORY · A PERSON · BALLPARK

SCENE 34W

Four seasons, one site, four plots each. WM-712 is ahead by eleven per cent in the two dry years and by two in the wet ones. The leaflet says eleven per cent higher yielding.

GUIDE 44W

Work out how many times bigger the dry-year gain is than the wet-year one. It is the bigger figure divided by the smaller one. One tile is the number of seasons, which is how much evidence there is rather than part of this sum.

ASK 12W

How many times bigger is the dry-year gain than the wet-year gain?

BEHIND THE BUTTON 110W

What a trial measures. Four plots per line over four seasons separate the line from the piece of ground it grew on. They do it under the weather those seasons happened to bring, and that weather is part of the answer.

Why the leaflet's number is a problem. Eleven per cent is true, and it is true of two dry years. A farmer in a wet season gets two, so a number quoted without its conditions is a different claim wearing the same digits.

What to write instead. The gain, and the seasons it belongs to, in one sentence. It is longer and it stays true whatever next summer does.

TAKES AS READ

a trial measures what happened under the conditions it happened in

VERDICT 143W

The two figures come from the same trial and the same four plots. In the two dry seasons WM-712 was eleven per cent ahead. In the wet ones it was two. How many times bigger is a division: $11 \div 2 = 5.5$. So the gain a farmer gets depends on the season more than five times over, and the leaflet prints the larger of the two numbers with nothing attached to it. Nobody made that number up, which is what makes it a hard problem: it is a true figure that becomes a false claim once the conditions are dropped. The sentence that survives says both parts at once — a big gain in dry seasons, a small one in wet, at this site. It is longer, it costs the headline, and it is still correct next summer whatever the weather does.

TAKEAWAY 18W

A number from a trial carries the conditions it was measured in, or it is a different claim.

3.3 What the crop cannot do without water

AGRONOMY & PLANT ENERGETICS · DRYING & PROCESSING HALL · A ROOM · CHAIN

SCENE 35W

Osman Toft, the agronomist, has eleven years of yield against summer rain, and a roughly straight line through it. A trainee asks why a plant built out of air minds a dry summer at all.

GUIDE 57W

Four steps carry carbon from the air into the grain. Each one has a number beside it saying how much it is passing today. Put them in the order the carbon travels, then name the step that is holding everything up. The biggest number on the board is not the answer, and neither is the first step.

ASK 19W

Trace the path carbon takes into the grain, and name the step holding it up in a dry summer.

BEHIND THE BUTTON 133W

What a governing step is. Along any path, the step passing the least decides the total. Improving any other step changes nothing at all, which is why naming the right one matters more than describing the path.

Why the pores are the awkward step. A leaf takes carbon dioxide in through tiny pores, and loses water out through the same pores. There is no way to open them for one and not the other. In a dry summer the plant shuts them to save water, and the carbon stops coming in.

What the other steps are doing. The leaves are catching as much light as ever, and the machinery inside them still works. The roots are moving water, not carbon. The number beside each step is what separates a suspect from the answer.

TAKES AS READ

plants take carbon dioxide in through tiny pores in their leaves

VERDICT 155W

Almost all the weight of a harvest arrives as carbon dioxide through tiny pores in the leaves. Water vapour leaves through those same pores. In a dry summer the plant shuts them, which is the right choice for staying alive and stops the carbon coming in at the same moment. Look at the numbers on the board. The leaves are catching light as well as ever, the machinery inside them could handle far more carbon than is arriving, and the roots are busy with water. Only one step has fallen, and it has fallen a long way. That is what a governing step means: the step passing the least decides the total, so improving any of the others changes nothing while the pores are shut. Toft's straight line is the same fact drawn over eleven summers, and it is why breeders talk about getting more grain per drop of water rather than about beating drought.

TAKEAWAY 15W

The opening a leaf takes carbon in through is the opening it loses water through.

3.4 Why the vault is cold

CONSERVATION & VIABILITY · SEED VAULT · A ROOM · CALLBACK · CHOICE

SCENE 34W

Mbaye has two lots of the same seed on the bench. One spent four years dried, at minus eighteen degrees. The other sat in a warm shed, and only three in ten still grow.

GUIDE 59W

Four accounts of why cold and dry works. Ask of each whether it treats the seed as alive or as dead. A dried seed is still slowly changing inside, so damage builds up over years. Two of these options have something being blocked or killed instead. And a clean sealed packet still lasts far longer when it is cold.

ASK 10W

Why does cold, dry storage keep a seed alive longer?

BEHIND THE BUTTON 169W

Why the wrong answers are worth reading. Each one is written to look right to somebody who made a particular mistake. Working out which mistake each one belongs to is most of the learning in this format — more than getting the right one, which you can sometimes do by feel.

Why sounding careful is not the same as being right. A long answer with a lot of detail in it feels safer, and it is easy to write one that is still wrong. The check that builds these questions measures option length across the whole game so the longest one is not the answer more often than chance. Reading is the only way through.

What the verdict adds. After you answer, the verdict says why each wrong option is wrong, in its own words. That is written for the reader who picked it. Read it even when you were right, because being right for a shaky reason is the thing that catches up with you two stops later.

TAKES AS READ

some seeds can survive being dried out and frozen

VERDICT 73W

A dried seed is still alive, and being alive means chemistry going on inside it. That chemistry slowly damages the parts the seed will need when it wakes up. Taking the water out slows those changes down. Cold slows them further, so the two work together. It is not a fuel tank running out. It is damage adding up, and everything in the vault is there to make it add up more slowly.

TAKEAWAY 14W

Keeping something alive for decades is a matter of slowing down what damages it.

A delay is not neutral, and neither is a release; the question is which cost is recoverable.

What comes with what you wanted

PLAN CARD 64W

Friday, and the plan is signed. WM-712 goes forward, forty of the sixty batches get regrown, and Qureshi has put in writing what she thinks of that. In the glasshouse, every line carrying the resistance gene is noticeably shorter than its sisters. Nobody chose that. Today you work out what came along with the gene, and whether the test can still be trusted here.

BRIEFING 11W

The resistant lines are shorter, and nobody bred them that way.

WHAT YOU DECIDE 11W

Find out what else the resistance gene is dragging behind it.

4.1 What came with the gene

HEREDITY & CROSSING · CROSSING HALL · A ROOM · CHOICE

SCENE 32W

Every line carrying the resistance is twelve to fifteen centimetres shorter than its sisters without it. The parent it came from was short. The parent it was bred back into is tall.

GUIDE 49W

Four explanations for shorter resistant plants. Ask of each whether it would affect every plant or only the ones carrying the new gene. Genes that sit close together on a chromosome tend to travel together. Splitting them needs a rare swap, which is why so many offspring get grown.

ASK 6W

Why are the resistant plants shorter?

BEHIND THE BUTTON 169W

Why the wrong answers are worth reading. Each one is written to look right to somebody who made a particular mistake. Working out which mistake each one belongs to is most of the learning in this format — more than getting the right one, which you can sometimes do by feel.

Why sounding careful is not the same as being right. A long answer with a lot of detail in it feels safer, and it is easy to write one that is still wrong. The check that builds these questions measures option length across the whole game so the longest one is not the answer more often than chance. Reading is the only way through.

What the verdict adds. After you answer, the verdict says why each wrong option is wrong, in its own words. That is written for the reader who picked it. Read it even when you were right, because being right for a shaky reason is the thing that catches up with you two stops later.

TAKES AS READ

genes sit in fixed places along a chromosome, in a fixed order

VERDICT 76W

Genes sit in a row along a chromosome, and a chromosome is handed on in stretches rather than gene by gene. So a gene you want arrives with whatever sits beside it. Here the resistance came from a short parent, and a piece of that parent's chromosome came with it. Splitting the two needs a swap in exactly that gap, which is rare. So you grow thousands, and look for the one plant where it happened.

TAKEAWAY 15W

A chromosome is passed on in stretches, so a wanted gene arrives with its neighbours.

4.2 A test fitted where it worked

MARKERS & GENE EXPRESSION · MOLECULAR LABORATORY · A PERSON · HOLDOUT

SCENE 72W

DNA is the set of instructions inside every seed. A marker is a short piece of DNA that sits near the gene you care about. Testing the marker is a cheap way to ask about the gene. This marker was worked out on the first population. Nakagawa now has a new cross, and a hundred plants scored both ways. A cut-off's score is how many of a hundred plants it calls right.

GUIDE 64W

Drag the cut-off and watch the score on the first population. Remember what that score is. The marker was worked out on these plants, so a cut-off tuned on them fits these plants as well as the trait. When you have picked one, press freeze. Freezing locks your cut-off and opens the new cross, which it has never seen. Report what it gets there.

ASK 26W

Set the cut-off on the plants the test was worked out on. Then freeze it and score it on the new cross. What does it get?

BEHIND THE BUTTON 168W

Why a marker works at all. The marker is not the gene. It is a piece of DNA close enough to the gene that the two are usually passed on together. So a plant with the marker usually has the gene. How well that holds depends on how close the two are in that family of plants.

Why a new parent changes the answer. Bring the trait in from a different parent and the DNA around the gene comes too. The marker and the gene may now sit further apart, or the marker may travel with the wrong copy of the gene. The marker did not change. The plants did.

Why you freeze before you look. A cut-off picked on the first population is flattered by it, because it was picked using those plants. The new cross is the honest test, and it is only a test while the cut-off was chosen without it. That is what pressing freeze means: you have decided, and now you find out.

TAKES AS READ

a test predicts a trait by sitting near the gene, in the population it was worked out on

VERDICT 69W

A cut-off chosen on one population partly fits that population's accidents, and accidents do not have to repeat. A broad shelf of good scores is worth trusting. It survives being frozen and used on a new cross. A sharp peak at one exact value does not. That peak was fitted to noise. Scoring on plants the rule has never seen is the only way to tell the two apart.

TAKEAWAY 19W

A rule fitted on one set of plants has to be scored on a set it has never seen.

4.3 Short is not simply worse

AGRONOMY & PLANT ENERGETICS · DRYING & PROCESSING HALL · A ROOM · BALLPARK

SCENE 32W

Toft points out that shorter wheats were the whole point of the 1960s. Less stem, more grain, and they stand up in the rain. Volpe still did not ask for fifteen centimetres.

GUIDE 57W

Work out what share of everything the plant built ended up in the grain. Take the grain weight and divide it by the whole weight the plant made, stem and roots and all. One tile is the weight of the stem on its own, which is part of the total rather than the thing you divide by.

ASK 12W

What share of everything the plant built ended up in the grain?

BEHIND THE BUTTON 131W

What the share means. A plant builds a fixed amount in a season, and then divides it between stem, leaves, roots and grain. The share that ends up in the grain is what a farmer takes away, and shorter wheats raised it by spending less on stem.

Why the stem is not waste. It holds the ear up, carries water and sugar to it, and keeps the crop off the ground in August rain. A stem strong enough to stand in weather is paid for out of the same total as the grain.

Why a shorter plant is not a better plant. It does not build more. It divides what it builds differently, which is a gain until the stem is too weak to hold the ear up in bad weather.

TAKES AS READ

a plant shares out what it builds between stem, leaf and grain

VERDICT 148W

A crop builds a certain amount in a season and then divides it up. This short wheat built 19.6 tonnes across the field, and 8.6 tonnes of that is grain: $8.6 \div 19.6 \approx 0.45$. A little under half. The tall wheat beside it built about the same total and got 0.33 of it into the ear, because more went into stem. That is the whole gain of the 1960s, and it is a gain in sharing rather than in building — neither plant photosynthesises faster than the other. It is also why the trade has a limit. The stem holds the ear up, carries water and sugar to it, and keeps the crop out of the mud in August rain, and all of that is bought out of the same total. Volpe's fifteen centimetres is not a rounding error either: it is stem, paid for with grain.

TAKEAWAY 16W

What matters is the share of what a plant built that ends up in the grain.

4.4 Why the same line is sown four times

FIELD TRIALS & SELECTION · FIELD LABORATORY · A ROOM · CALLBACK · CHOICE

SCENE 31W

Meera Quiroga has the plot map on the wall. Two hundred and forty plots, each line sown four times across four blocks, in scrambled order. The north end is visibly wetter.

GUIDE 52W

Four things four plots might buy. Ask of each whether it is about the comparison or about safety. A field is not the same everywhere: drainage, depth and shelter all vary. So one plot's yield is the seed and the ground together. Spreading a line across the field is what separates them.

ASK 10W

What does sowing each line in four scattered plots buy?

BEHIND THE BUTTON 169W

Why the wrong answers are worth reading. Each one is written to look right to somebody who made a particular mistake. Working out which mistake each one belongs to is most of the learning in this format — more than getting the right one, which you can sometimes do by feel.

Why sounding careful is not the same as being right. A long answer with a lot of detail in it feels safer, and it is easy to write one that is still wrong. The check that builds these questions measures option length across the whole game so the longest one is not the answer more often than chance. Reading is the only way through.

What the verdict adds. After you answer, the verdict says why each wrong option is wrong, in its own words. That is written for the reader who picked it. Read it even when you were right, because being right for a shaky reason is the thing that catches up with you two stops later.

TAKES AS READ

the same crop grows differently in different parts of a field

VERDICT 76W

Ground varies. Drainage, depth, shelter, how hard it was driven over. Sow a line once and its yield is the seed and that patch mixed together, with no way to tell them apart. Sow it four times in different blocks and the patches differ while the seed does not. Scrambling the order stops anybody's habits handing good ground to a favourite. How much the four disagree is itself a measure of how much the ground matters.

TAKEAWAY 19W

Repeating a thing in different places is how you find out which part of the answer was the place.

Genes near each other are inherited together, which is what makes markers work and what makes them expensive.

How many is enough

BEFORE THE DAY — HUNT

Six bagged heads, and the sheet lists four

Every bagged head has to be found before the wind gets up. A bag that has come off means a cross with an unknown father, and nothing afterwards can tell you which one it was. Four are where the sheet says. Find the rest.

PLAN CARD 73W

Monday of the second week, and the regrow list has to be cut from sixty to forty. Tomas Ivanescu has worked out how many plants each regrow needs, and it is not the number in the manual. Mbaye has eleven years of tests on one batch, and a line the results are about to cross. Today you size the samples, and work out which of last year's results were really the ground talking.

BRIEFING 11W

40 accessions, 1 field, and a number nobody wants to hear.

WHAT YOU DECIDE 15W

Put a number on what a regeneration needs, and on what the vault is losing.

5.1 Not the number of plants

POPULATION GENETICS · GENETIC RESOURCES OFFICE · A ROOM · BALLPARK

SCENE 31W

Last year's grow-out used a hundred and twenty plants. Only eighteen of them gave pollen, and a hundred and two set seed. Ivanescu wants that looked at before the next plan.

GUIDE 53W

Four numbers, and two of them belong to other questions. The seed plants, and half the plot. Ask which two a share needs. A share compares one group with the whole, so the total is the plants standing rather than any part of it. The pollen side is much the smaller group here.

ASK 7W

What share of the plants gave pollen?

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

this crop keeps pollen and seed on separate plants · every seed has one parent on each side

VERDICT 72W

Every seed gets half its genes from a pollen plant and half from a seed plant. So the pollen side is a doorway that everything has to come through. Here that doorway was eighteen plants wide, out of a hundred and twenty standing in the plot. Fifteen per cent. What it costs is variety. Whatever those eighteen plants did not carry is gone from the batch, however many plants were standing there.

TAKEAWAY 20W

A crowd of plants can pass on as little as a handful, if one side of the family is small.

5.2 Write the rule before it falls

CONSERVATION & VIABILITY · SEED VAULT · A PERSON · TRIGGER

SCENE 33W

Mbaye has eleven germination tests on one batch, falling from ninety-eight per cent to eighty-five. The rule says grow it out at eighty-five. Booking a field takes time, and so does the grow-out.

GUIDE 72W

One rule, written before the results arrive. Once you release it the number cannot change. The grow-out takes one reading to get going from the moment it is called. Below 85 % germination the lot is past the floor. So your line has to sit above 85 by however far the number falls in one reading — and not so far above it that a healthy lot is grown out for nothing.

ASK 16W

At what germination do you start the grow-out, when it takes one reading to get going?

BEHIND THE BUTTON 123W

Why a floor is not a trigger. The floor is where the seed must already have been grown out and replaced. Everything that has to happen first takes weeks, so the trigger has to sit above the floor by however long the work takes.

What the falling numbers tell you. Eleven tests falling from ninety-eight to eighty-five is a trend, not scatter. How fast it is falling is what turns a threshold into a date, and a rule written against the level alone ignores that.

Why the rules go first. Test results arrive one at a time. A threshold moved as they arrive is just a description of what somebody already decided, and the next person to run the vault cannot follow it.

TAKES AS READ

germination is the share of seeds that actually grow · a grow-out needs field space booked before it can start

VERDICT 76W

The floor is where the seed stops being good enough. It is the wrong moment to start planning, because everything you would do next takes time. Booking a field takes several updates. Starting the grow-out takes another. Set the thresholds before the numbers arrive, and hindsight cannot quietly move them later. Reserve at ninety and start at eighty-six, and both land in time. Wait for eighty-five and the work begins after the line has been crossed.

TAKEAWAY 16W

A threshold has to fire early enough for the thing it starts to arrive in time.

5.3 A difference that was the field

FIELD TRIALS & SELECTION · FIELD LABORATORY · A ROOM · VERIFY

SCENE 34W

Last season two pure lines differed by nine centimetres in height, clearly. This season the same lines went onto different plots, and now they do not differ at all. The seed has not changed.

GUIDE 47W

Say what the height difference will be this year before the plots are measured. Write a number down first. If last year's nine centimetres was the seed, it comes back wherever the lines are sown. If it was the ground, it goes away when the ground changes.

ASK 13W

Predict this year's height difference between the two lines, then measure the plots.

BEHIND THE BUTTON 119W

Why a prediction first. Any result can be explained afterwards. A number written before the measuring is a test of the explanation, and the two explanations here predict different numbers.

What the two explanations are. Either the lines really do differ in height, in which case the difference travels with the seed, or last year's plots differed — deeper soil, more water, less wind — in which case the difference stays with the plots.

Why the same line is sown four times. Four plots per line, spread around the field, so that no line's score is really a score for one corner of the ground. It is the same idea as taking an average instead of measuring one plant.

TAKES AS READ

the same line grown in two places can give two different answers

VERDICT 142W

Two explanations were available and they predicted different numbers. If the two lines really differ in height, the difference travels with the seed and shows up again wherever they are sown. If last year's plots differed — deeper soil at that end, more water, some shelter — the difference stays behind with the plots. Measured this season, on new ground, the lines differ by about a centimetre, which is nothing. So last year's nine centimetres was the field. Nobody did anything wrong last season: the plots were measured correctly and the number was real. It was just a number about a piece of ground rather than about a kind of wheat. That is what sowing each line in four scattered plots is for, and why a result that does not survive being grown somewhere else is treated as being about the somewhere.

TAKEAWAY 15W

A result that does not survive being grown somewhere else was probably about the somewhere.

5.4 Two parents, and what you are asking of them

HEREDITY & CROSSING · CROSSING HALL · A ROOM · CALLBACK · CHOICE

SCENE 34W

Volpe has two parents pinned above the bench. One is a modern wheat that yields well. The other is an old local kind that came through the 2018 drought. Bengtsson asks what she wants.

GUIDE 57W

Four accounts of what a cross is for. Ask of each whether it could happen when each parent passes on half its genes. The offspring get a mix, not an average. Neither parent already has the combination being looked for. And many of the offspring will be worse than both, because the bad genes get shuffled too.

ASK 11W

What is the programme actually asking for by crossing these two?

BEHIND THE BUTTON 169W

Why the wrong answers are worth reading. Each one is written to look right to somebody who made a particular mistake. Working out which mistake each one belongs to is most of the learning in this format — more than getting the right one, which you can sometimes do by feel.

Why sounding careful is not the same as being right. A long answer with a lot of detail in it feels safer, and it is easy to write one that is still wrong. The check that builds these questions measures option length across the whole game so the longest one is not the answer more often than chance. Reading is the only way through.

What the verdict adds. After you answer, the verdict says why each wrong option is wrong, in its own words. That is written for the reader who picked it. Read it even when you were right, because being right for a shaky reason is the thing that catches up with you two stops later.

TAKES AS READ

young plants get half their genes from each parent

VERDICT 74W

Each parent hands on half its genes, so the first generation gets one set from each. The generation after that shuffles those sets into thousands of combinations. What the breeder wants is the yield genes of the modern parent together with the drought genes of the old one. Neither parent has that. Most of the offspring will be worse than either. Breeding is growing enough of them to find the rare ones worth keeping.

TAKEAWAY 19W

A cross is a request for a combination, and it takes generations to find out whether it was granted.

END OF THE DAY 17W

Every number here is about a population, and populations do not behave like the plants in them.

What you asked for and what you got

BEFORE THE DAY — CANVASS

Which growers kept the seed apart

Growers took seed from Wellmere last year and agreed to keep it away from other wheat. What the collection can say about those lines depends on whether they did. Ask round the farms until enough answers have come back.

PLAN CARD 73W

Harvest week, and the trial is scored. Volpe has picked the best twenty plots out of two hundred and forty, and wants them all kept. Quiroga says most of that head start is the ground they stood on, and will not travel to next year. Two of the best plots also covered the ground eleven days early. Today you work out what the choosing really buys, and why those two did so well.

BRIEFING 11W

The best 20 plots, and what their offspring will be like.

WHAT YOU DECIDE 9W

Predict what one round of selection will actually deliver.

6.1 What one round returns

FIELD TRIALS & SELECTION · FIELD LABORATORY · A ROOM · BALLPARK

SCENE 28W

The trial averages 7.4 tonnes a hectare. The twenty plots Volpe wants to keep average 9.1. Quiroga says about a third of that head start gets passed on.

GUIDE 52W

Five numbers, and two belong elsewhere: the number of plots, and a head start already worked out for you. Ask what this sum needs. The chosen plots were ahead for two reasons, and only one of them gets passed on. So their own average is not what the next generation will do.

ASK 6W

What should the next generation average?

BEHIND THE BUTTON 106W

Why an estimate rather than a calculation. Every quantity on this board is already measured or stated, so nothing here has to be derived. What is being tested is whether you can pick the quantities the relationship actually needs, and whether the size of the answer looks right once they are in place.

Why the tiles carry labels. A bare number cannot be checked against the relationship it is going into. Reading the label is how a unit mismatch, or a quantity belonging to a different part of the problem, is caught before it is placed — which is the habit this format exists to build.

TAKES AS READ

a plant's performance is partly what it inherited and partly where it grew

VERDICT 77W

The plots you choose are above average for two reasons. Some of it is the plants, and some of it is the ground they happened to stand on. Only the first travels to the next generation. Here the head start is 1.7 tonnes, and about three tenths of that is inherited, so the gain is roughly half a tonne. Expect 9.1 next year and the disappointment is guaranteed. Expect 7.9 and the programme is honest with itself.

TAKEAWAY 17W

Only the inherited share of an advantage is passed on, and for yield that share is small.

6.2 What a season costs

HEREDITY & CROSSING · CROSSING HALL · A PERSON · CHOICE

SCENE 32W

Volpe has the pipeline on the board. Cross to release is seven years, and everything behind WM-712 moves back with it. Qureshi asks what a year of a wheat variety is worth.

GUIDE 57W

Four accounts of what a season costs. Ask of each whether it is about this line or about everything behind it. The programme shares seasons, ground and people. Holding one line back holds back its seed increase, the trial that would have used the ground, and the crosses waiting. That lands on lines nobody was arguing about.

ASK 11W

Why does holding it one season cost more than one season?

BEHIND THE BUTTON 169W

Why the wrong answers are worth reading. Each one is written to look right to somebody who made a particular mistake. Working out which mistake each one belongs to is most of the learning in this format — more than getting the right one, which you can sometimes do by feel.

Why sounding careful is not the same as being right. A long answer with a lot of detail in it feels safer, and it is easy to write one that is still wrong. The check that builds these questions measures option length across the whole game so the longest one is not the answer more often than chance. Reading is the only way through.

What the verdict adds. After you answer, the verdict says why each wrong option is wrong, in its own words. That is written for the reader who picked it. Read it even when you were right, because being right for a shaky reason is the thing that catches up with you two stops later.

TAKES AS READ

a breeding programme moves one generation at a time

VERDICT 73W

A breeding programme is a queue sharing the same seasons, the same ground and the same people. Hold this line for a year and its seed increase waits. So does the trial that would have used that ground, and the crosses behind it. The bill lands on lines that were never part of the argument. That does not settle whether releasing is right. It makes the price of waiting something everybody can see.

TAKEAWAY 15W

The cost of holding something back is rarely paid by the thing being held back.

6.3 Why those two did well

AGRONOMY & PLANT ENERGETICS · DRYING & PROCESSING HALL · A ROOM · CONTROL

SCENE 34W

Two of the twenty chosen plots covered the ground eleven days before the rest and flowered four days earlier. Their yields are the best in the trial, and the season went dry after flowering.

GUIDE 61W

Four things are different about those two plots, and the yield sheet will give you a figure for whatever you set up. Change one of them and grow it. Change two and the figure cannot say which one did it. Then set your one back the way it was, because a difference that does not come back was never the reason.

ASK 10W

Which difference between those plots is actually raising the yield?

BEHIND THE BUTTON 120W

Why one thing at a time. Any two plots differ in dozens of ways at once. A test that changes several of them gives a number that fits every story, which is the same as having no answer.

What covering the ground early buys. Leaves catch light. Covering the ground eleven days sooner means eleven more days of catching it while there is still water in the soil to use it with, and this season went dry after flowering.

Why setting it back matters. If the yield falls again when the early cover goes, the early cover was doing it. If it stays up, something else was, and the test has saved a season of breeding for the wrong thing.

TAKES AS READ

a crop builds most of its grain from light caught after it flowers

VERDICT 146W

Four things were different about the best two plots. Only one of them holds up when it is tested on its own. Sow plots that cover the ground eleven days early. Keep the seed, the date, the water and the nitrogen the same. The yield goes from 6.2 tonnes to 8.3. Take the early cover away and it drops back to 6.4. That is a real cause. The figure follows the change both ways, and it moves far more than the trial's own wobble. The others do nothing on their own. Flowering four days early is worth less than the wobble. So is where the plot sits, and so is the extra nitrogen. Leaves catch light. This season went dry after flowering. So eleven extra days of catching light, while there was still water to use it with, is eleven days the other plots never got.

TAKEAWAY 14W

An early crop catches light while there is still water to use it with.

6.4 What the screen may not find

MARKERS & GENE EXPRESSION · MOLECULAR LABORATORY · A ROOM · CALLBACK · CHOICE

SCENE 34W

Nakagawa is screening fifty batches for a second resistance gene. Twenty-two of them are on the drifted list, and their test controls are clean. She wants to know what a clean result still means.

GUIDE 55W

Four readings of what drift does to a test. Ask of each whether it changes clean results, positive results, or both. Chance can remove a gene that was rare, so a clean result now has two stories and the test cannot tell them apart. Chance cannot invent the gene, so a positive still means something.

ASK 8W

What does the drift mean for the screen?

BEHIND THE BUTTON 169W

Why the wrong answers are worth reading. Each one is written to look right to somebody who made a particular mistake. Working out which mistake each one belongs to is most of the learning in this format — more than getting the right one, which you can sometimes do by feel.

Why sounding careful is not the same as being right. A long answer with a lot of detail in it feels safer, and it is easy to write one that is still wrong. The check that builds these questions measures option length across the whole game so the longest one is not the answer more often than chance. Reading is the only way through.

What the verdict adds. After you answer, the verdict says why each wrong option is wrong, in its own words. That is written for the reader who picked it. Read it even when you were right, because being right for a shaky reason is the thing that catches up with you two stops later.

TAKES AS READ

a gene that is rare in a population may be missing from a small sample of it

VERDICT 79W

Drift can remove a gene that was rare to begin with. So a clean result from one of those batches has two explanations. The gene was never there, or it was there and got lost in a grow-out. The test cannot tell those apart, because it is looking at the seed in front of it. A positive is still worth having, because drift removes genes and never invents them. The fix is to test material that missed the bottleneck.

TAKEAWAY 22W

A screen finds what is in front of it, and what is in front of it is what survived the last grow-out.

END OF THE DAY 18W

What a round of selection returns is the part of the difference that was inherited, and no more.

The collection is not what the records say

PLAN CARD 84W

Thursday. Nakagawa ran a routine DNA test across ninety batches. DNA is the instruction set inside every seed, and it can be read directly. A third of them do not match their records — same numbers, same paperwork, different plants. Sylvie Pirelli has checked the labels twice and found nothing wrong. Qureshi has gone down to the vault and has not come back up. Today you work out what happened to those batches, and whether it happened in the vault or in the field.

BRIEFING 9W

A genotyping run that was meant to be routine.

WHAT YOU DECIDE 12W

Find out why the regenerated accessions no longer match their own passports.

7.1 Which evidence shares the grow-out

POPULATION GENETICS · GENETIC RESOURCES OFFICE · A ROOM · TRACE

SCENE 26W

Thirty of ninety batches no longer match their records from the 1990s. Nakagawa has four kinds of evidence on the board. The shelf checks are clean.

GUIDE 54W

Open each kind of evidence to see what it was worked out from. Keep the ones that stand apart from the repeated small grow-outs, and untick the ones that inherit them. Then name what the failing ones share. The shelf checks being clean is itself evidence about where the drift did not come from.

ASK 13W

Which results inherit the repeated small grow-outs, and which stand apart from them?

BEHIND THE BUTTON 122W

What repeated small grow-outs do. Growing a batch back from only a few plants loses some of its variety by chance, and doing that again and again keeps losing more. The result looks exactly like a collection that has drifted away from its old records — because it has, through its own upkeep.

Why clean shelves matter. If the storage or the labelling were at fault, the shelf checks would show it. Clean shelves point away from an accident and towards something in how the seed has been renewed.

Why some evidence survives. Germination tests and the vault log record how seed was handled, not what genes it carries. That is why they can test the grow-out history instead of repeating it.

TAKES AS READ

a result can depend on how something was handled, as well as on today's test · evidence is only independent if it does not share the important step

VERDICT 79W

Three of the four kinds of evidence rest on the same grow-out history, so their agreement is one cause counted three times. The germination evidence does not. It rests on tests of whether the seed grows, and it says the seed is fine. Opening what each one depends on splits the story in two. The store kept the seed alive. The grow-outs changed what was in it. Throwing out the germination evidence would lose the half that stands alone.

TAKEAWAY 15W

Several results can agree because they share one history, and that is not several reasons.

7.2 The vial nobody has grown

MARKERS & GENE EXPRESSION · MOLECULAR LABORATORY · A ROOM · CHOICE

SCENE 37W

Two of the fifty batches carry the resistance gene, and the screenhouse can grow only one this season. The full vial has been grown out three times. The thin one has never been grown out at all.

GUIDE 54W

Four options, and two facts to weigh: how much seed there is, and how many times it has been grown out. Ask of each which of the two it uses. The amount in the vial says nothing about what is in it. Every grow-out is another chance for rare things to be left out.

ASK 25W

One vial is full and has been grown out three times. The other is thin and never grown. Which is closer to the original plant?

BEHIND THE BUTTON 169W

Why the wrong answers are worth reading. Each one is written to look right to somebody who made a particular mistake. Working out which mistake each one belongs to is most of the learning in this format — more than getting the right one, which you can sometimes do by feel.

Why sounding careful is not the same as being right. A long answer with a lot of detail in it feels safer, and it is easy to write one that is still wrong. The check that builds these questions measures option length across the whole game so the longest one is not the answer more often than chance. Reading is the only way through.

What the verdict adds. After you answer, the verdict says why each wrong option is wrong, in its own words. That is written for the reader who picked it. Read it even when you were right, because being right for a shaky reason is the thing that catches up with you two stops later.

TAKES AS READ

every grow-out is a chance for a batch to drift from what it was · a small lot of living seed can usually be built back up

VERDICT 97W

A seed batch is not a photograph. Every time it is grown out, only some of the plants set seed, and the batch that comes back is slightly different from the batch that went in. Do that three times and the difference adds up. The thin vial has never been through that, so whatever was collected is still in it. Being thin is a problem you can fix — three hundred seeds carefully grown is a full vial again. Being drifted is not a problem you can fix, because there is nothing left to go back to.

TAKEAWAY 18W

Copies of copies drift. The one nobody has copied is the one that still holds what was collected.

7.3 What the store did right

CONSERVATION & VIABILITY · SEED VAULT · A ROOM · CHAIN

SCENE 29W

Qureshi wants one sentence for the board. It has to name what failed, rather than the room it happened near. The vault log covers twenty-eight years with no fault.

GUIDE 49W

Follow the seed's path from the freezer back into the store, and name the step where it lost what it lost. Each step has a note saying how well it did its job. The freezer is the biggest thing on the board and it is not automatically the answer.

ASK 15W

Follow the seed's path and name the step where the batch lost what it lost.

BEHIND THE BUTTON 114W

What the freezer does. Cold keeps seed alive for longer. The log covers twenty-eight years with no fault, and seed from the affected packets still grows well, which is what a working freezer looks like from outside.

What a grow-out is. Growing a batch out to make fresh seed. It is a sampling step, so how many plants went into it decides how much of the batch came back.

Why naming the step matters. The fix follows the step. Call this a storage failure and the money goes to a freezer that worked; call it a grow-out failure and the money goes to plant counts and pollen distance, which is where it was lost.

TAKES AS READ

a store keeps what it is given rather than improving it

VERDICT 138W

A store keeps what it is handed. The freezer log covers twenty-eight years without a fault and seed from the affected packets still grows well, so the cold end of this path did its job. The step that did not is the grow-out. Growing a batch out to make fresh seed means taking a sample of it, and this one used thirty plants where two hundred were needed. Whatever those thirty did not carry was gone before the packets went back on the shelf, and no freezer restores it. The label and the record are part of the seed, which is why this matters for the sentence Qureshi wants: naming the room sends money to equipment that worked. Naming the step sends it to plant counts and pollen distance, which is where the collection is actually losing things.

TAKEAWAY 17W

Storage held what it was handed. The loss happened in the field, on the way back in.

7.4 Gene present, resistance working?

MARKERS & GENE EXPRESSION · MOLECULAR LABORATORY · A PERSON · CHOICE

SCENE 31W

Nakagawa can run the DNA test on two-leaf seedlings. On final plants she can also check whether the gene is switched on after rust, and put real rust on the leaves.

GUIDE 51W

Four ways to order the tests. Ask of each what a DNA test can and cannot show. Carrying a gene is not the same as using it. Between the gene and a plant that survives rust sit several steps, and only one kind of test asks whether the whole chain works.

ASK 13W

The DNA test says the gene is there. What has it not shown?

BEHIND THE BUTTON 169W

Why the wrong answers are worth reading. Each one is written to look right to somebody who made a particular mistake. Working out which mistake each one belongs to is most of the learning in this format — more than getting the right one, which you can sometimes do by feel.

Why sounding careful is not the same as being right. A long answer with a lot of detail in it feels safer, and it is easy to write one that is still wrong. The check that builds these questions measures option length across the whole game so the longest one is not the answer more often than chance. Reading is the only way through.

What the verdict adds. After you answer, the verdict says why each wrong option is wrong, in its own words. That is written for the reader who picked it. Read it even when you were right, because being right for a shaky reason is the thing that catches up with you two stops later.

TAKES AS READ

a nearby marker can come apart from the gene it stands for · a gene has to be switched on before it does anything

VERDICT 69W

The DNA test is worth running early, because it clears out most seedlings months before rust could be used. It is not proof. The marker sits near the gene rather than inside it, and the two can come apart. Even the gene itself only says what the plant carries. It has to switch that gene on when rust arrives. So the last plants get real rust put on them.

TAKEAWAY 14W

DNA says what a plant carries. Only a real test says whether it works.

7.5 What was lost, and when

POPULATION GENETICS · GENETIC RESOURCES OFFICE · A PERSON · BALLPARK

SCENE 31W

Ivanescu has the 1994 field notes for the drifted batch. The collector saw rust on the crop, and noted that about one plant in twenty stayed clean. Today's sample has none.

GUIDE 54W

Work out how many resistant plants a grow-out of thirty should have picked up. Five plants in every hundred carried the resistance, so take the thirty plants, multiply by five, and share the answer across the hundred. One tile is the two hundred plants the standard asks for, which is the number nobody used.

ASK 12W

How many resistant plants should a grow-out of thirty have picked up?

BEHIND THE BUTTON 119W

What one in twenty means. Five plants in every hundred carried the resistance in 1994. It was there, and it was rare, and rare is the whole problem.

What a grow-out of thirty does. It takes thirty plants and lets them make the next batch. With one in twenty carrying the gene, thirty plants are expected to include one or two — and quite often none at all, the way five coin tosses sometimes give no heads.

Why nothing notices. The seed that comes back germinates well, looks right and passes every routine check. What is missing is a gene nobody tested for, and no freezer puts it back once none of the plants that went in carried it.

TAKES AS READ

something rare in a population can be missed altogether when a few plants are taken

VERDICT 139W

In 1994 five plants in every hundred were resistant, which is one in twenty. A grow-out of thirty plants is therefore expected to include $30 \times 5 \div 100 = 1.5$ of them — one or two, and often none, in the same way five coin tosses sometimes give no heads. So the gene did not have to be destroyed to disappear. It only had to be missed. Two hundred plants would have picked up about ten of them, which is why the standard says two hundred and why the missing plant count on the 1994 sheet matters so much. And nothing about the batch that came back looks wrong: it germinates well, it looks right, and it passes every routine check, because what is gone is a gene nobody tests for until a rust arrives that needs it.

TAKEAWAY 17W

A gene that has vanished was not necessarily destroyed. It may only have failed to be picked.

7.6 The copy that was never in the vault

CONSERVATION & VIABILITY · SEED VAULT · A PERSON · CHOICE

SCENE 28W

Fenn's old wheats are growing in the station plots. His family has grown one of them without a break since before Wellmere existed. Rust arrives there most years.

GUIDE 49W

Four reasons the field copy might hold what the vault lost. Ask of each whether it names a reason or just a difference. Fresher and bigger are not reasons. Rust arriving every year means resistant plants leave more seed. In a clean nursery, resistance had no advantage at all.

ASK 11W

Why is Fenn's field likely to hold what the vault lost?

BEHIND THE BUTTON 169W

Why the wrong answers are worth reading. Each one is written to look right to somebody who made a particular mistake. Working out which mistake each one belongs to is most of the learning in this format — more than getting the right one, which you can sometimes do by feel.

Why sounding careful is not the same as being right. A long answer with a lot of detail in it feels safer, and it is easy to write one that is still wrong. The check that builds these questions measures option length across the whole game so the longest one is not the answer more often than chance. Reading is the only way through.

What the verdict adds. After you answer, the verdict says why each wrong option is wrong, in its own words. That is written for the reader who picked it. Read it even when you were right, because being right for a shaky reason is the thing that catches up with you two stops later.

TAKES AS READ

a crop growing where a disease turns up is being tested by it every year

VERDICT 75W

Rust has come through Fenn's field most years. In a rust year the plants that shrug it off set more seed, and their share of the next crop goes up. Nobody planned that. It happens because the disease is there. The vault's grow-outs were done in clean nurseries, where resistance bought a plant nothing, so anything rare could quietly vanish. His field is not a frozen copy of 1994. It may be better than one.

TAKEAWAY 19W

A population kept in the ground keeps being tested by the thing you might one day need it for.

7.7 Covered early, or open to the wind

AGRONOMY & PLANT ENERGETICS · DRYING & PROCESSING HALL · A ROOM · BELT

SCENE 29W

Two hours of bags have to be cleared. A head that was bagged before it flowered could only pollinate itself. A head left open took whatever the wind brought.

GUIDE 42W

Two bins. Ask when the cover went on. Before the flowers opened, the plant pollinated itself and the line stays the same. Open at flowering, and the father could be any plant upwind. The words on the label will not tell you.

ASK 12W

Send each bag to the bin that says what fathered its seed.

BEHIND THE BUTTON 64W

Why the date decides it. A cover put on before flowering keeps other pollen out. A cover put on afterwards keeps nothing out, because the pollination has already happened.

Why it matters at the store door. Seed from a selfed head grows into the same line again. Seed with an unknown father is something else, and it looks exactly the same in the bag.

TAKES AS READ

a flower can be pollinated by its own plant or by another one

VERDICT 108W

One moment decides which bin a bag belongs in. Was the head covered before its flowers opened? If it was, the only pollen that could reach it was its own, and the seed grows into the same line next year. If it was open when it flowered, pollen came from whatever was flowering upwind, and here that means the trial plots eighty metres away. A cover put on afterwards changes nothing at all. That is why the labels are no help and are not meant to be. Covers, sleeves and sheets turn up on both sides of the floor. The date is the category, and nothing else is.

TAKEAWAY 10W

What fathered a seed was decided before the flower opened.

7.8 Counting a generation

HEREDITY & CROSSING · CROSSING HALL · A ROOM · CALLBACK · BALLPARK

SCENE 29W

Four hundred plants from the second generation. Three hundred and six shrug the disease off, and ninety-four do not. The breeder asks what that number was supposed to be.

GUIDE 55W

Work out what number was expected before looking at what came. Each parent carries one resistant copy and one that is not, so three plants in four should shrug the disease off. That is a share of the four hundred. One tile is the count that actually came, which is what you are comparing against.

ASK 12W

How many of the four hundred plants should shrug the disease off?

BEHIND THE BUTTON 118W

Where three in four comes from. Each parent gives one of its two copies to each seed. There are four ways the copies can pair up, and three of those four contain a resistant copy. One resistant copy is enough, so three in four look resistant.

Why the whole generation is the evidence. Any single plant is resistant or not, and tells you nothing. Four hundred of them make a proportion, and a proportion is what a breeding rule predicts.

Why close is the right answer. Nobody expects exactly three hundred. Coin tosses do not come out exactly half, and four hundred plants is enough for the count to land near the expected number rather than on it.

TAKES AS READ

a plant carries two copies of each gene and passes on one of them

VERDICT 142W

Each parent has two copies of the gene and gives one to each seed. There are four ways those copies can pair up, and three of the four include a resistant copy. One copy is enough to shrug the disease off, so three plants in four should manage it. Three quarters of four hundred is $400 \times 0.75 = 300$. The tray gave 306, which is as close as anybody should expect — a proportion lands near its expected number, not exactly on it, in the same way a hundred coin tosses rarely give exactly fifty heads. That is what makes the count evidence. If two genes were doing the work, and either one was enough, the expected number would be 375 instead, and 306 is nowhere near it. No single plant in the tray could have told you any of this.

TAKEAWAY 13W

The proportions across a whole generation carry information that no single plant does.

END OF THE DAY 14W

A collection can be kept perfectly and still stop being the thing it was.

What goes in the crossing block

BEFORE THE DAY — EVADE

The merchant who wants a yes at the gate

The merchant has been on the causeway since eight, wanting an answer about WM-712 before the rust question is settled. Nothing is agreed outside the office. Keep out of his way and finish your round.

PLAN CARD 66W

Thursday. The crossing block goes in tomorrow morning, and it decides what the programme works on until 2033. A cross made in it is a line in four years and a release in seven. Volpe has her list and Qureshi has hers, and this year they overlap. There is room for eight crosses. Today you fix the eight. After tomorrow nothing this season can be changed.

BRIEFING 11W

The block is drilled tomorrow. After that the parents are fixed.

WHAT YOU DECIDE 17W

Decide what the next seven years are made of, on the last day it can be changed.

8.1 Eight crossing commitments

HEREDITY & CROSSING · CROSSING HALL · A ROOM · ALLOCATE

SCENE 29W

The crossing board has eight slots for the next two years. Volpe wants yield crosses. Qureshi wants new resistance brought in. The elite line still needs the second gene.

GUIDE 52W

Eight slots for two years, and more requests than slots. Watch the list of questions rather than the board: it shows what the programme can still answer with your plan. Every package you commit is a question answered and one given up, and one request can wait a season without costing anything.

ASK 15W

Which packages keep the programme defended without spending the board on work that can wait?

BEHIND THE BUTTON 110W

What defended means here. One resistance gene gets beaten eventually. A line carrying two takes far longer to beat. Getting the second gene into the best line is what keeps everything else the programme has built worth using.

Why new resistance is worth slots at all. Bringing resistance in from an unrelated plant is slow and produces nothing sellable for years. It is also the only way to have a third gene ready when the second one fails.

Why some work can wait. A question whose answer will be just as useful next season is not competing with one that has to be answered before the elite line is released.

TAKES AS READ

a cross ties up glasshouse space and staff for several seasons · closely related parents share the same weaknesses

VERDICT 69W

There are two urgent questions and eight slots. One is getting the second resistance gene into the wheat farmers already grow. The other is getting it into unrelated wheat, so the programme does not rebuild the same weak point. Yield crosses between elite parents are useful, and they narrow the same base further. Fenn's plants will not be understood for two seasons. Spend the board on what cannot wait.

TAKEAWAY 18W

A breeding plan is the set of questions it chooses to answer, and every slot spent delays another.

8.2 How narrow the base has become

POPULATION GENETICS · GENETIC RESOURCES OFFICE · A ROOM · VERIFY

SCENE 32W

Ivanescu has six elite parents' family trees across the wall. Four share a recent grandparent. All six carry the same resistance gene, and the new rust is beating it east of here.

GUIDE 50W

Say how many of the six lines will lose to the new rust, then infect all six and count. Write the number down first. Four of them come down from one grandparent and carry that plant's resistance gene, and the new rust is already beating that gene east of here.

ASK 17W

Predict how many of the six lines the new rust gets past, then infect them and count.

BEHIND THE BUTTON 109W

Why the family tree matters. Related parents share more of their genes than unrelated ones do. Four of these six share a grandparent, so a gene that came from that grandparent is in all four of them.

What one gene beaten means. A rust race that gets past a gene gets past it in every plant carrying it, on the same afternoon. It has to be beaten once, not four times.

Why the other two are worth so much. Their resistance came from somewhere else, so a rust that beats the first gene has said nothing about theirs. Two different sources are worth more than six copies of one.

TAKES AS READ

related parents share more of their genes than unrelated ones do

VERDICT 155W

Four of the six parents come down from the same 1998 grandparent, and the resistance all four carry is that one plant's gene. The new rust has already got past that gene east of here. So it does not have to beat four defences; it has to beat one, and four lines fall together. Infected in the screenhouse, that is exactly what happens: four go down and two hold. The two that hold got their resistance from somewhere else — one from a cross with a Chinese wheat, one from an Anatolian landrace — and a rust that beats the first gene has said nothing about either of those. That is what a narrowing base costs, and it is why the pedigree board is worth having on the wall. Every line in the nursery scoring resistant looked like strength. Once you know where the resistance came from, it looks like one gene counted six times.

TAKEAWAY 18W

A programme that keeps crossing its best with its best is narrowing what it can still draw on.

8.3 One disagreement, two possible causes

MARKERS & GENE EXPRESSION · MOLECULAR LABORATORY · A ROOM · DEGENERACY

SCENE 32W

Nakagawa has four hundred plants scored two ways: by a DNA test, and by whether they fell ill. Eleven carry the marker and still fell ill. The test's own controls look clean.

GUIDE 53W

Two controls slide the amount of swapping and the amount of test error along a family of mixes that all give eleven disagreements in four hundred plants. See how wide that family is before naming a cause. Then bring in the direct gene test, which fails for one explanation and not the other.

ASK 18W

Which mixes of swapping and test error fit the disagreement, and which one survives the direct gene test?

BEHIND THE BUTTON 131W

Why the disagreement rate alone cannot decide. The marker sits near the gene but not on it, so sometimes they are separated when plants are made. A test that occasionally misreads produces the same number of disagreements. Eleven in four hundred fits several mixtures of the two.

What the direct gene test does. It asks about the gene itself instead of the marker nearby, so swapping cannot affect it. A plant that has the gene and still fell ill is a test error. A plant that does not have the gene was a swap.

Why clean controls are not enough. Controls show the test worked on known material. They cannot catch a failure in one particular plant's sample, which is exactly what produces a handful of odd results in four hundred.

TAKES AS READ

a DNA test looks for a chosen stretch of DNA · a marker sits near a gene rather than inside it · a plant swaps pieces between its two copies of a chromosome when it makes seed

VERDICT 77W

The DNA test here does not look at the resistance gene. It looks at a marker sitting near it. Near is not inside. When a plant makes seed, its two copies of a chromosome swap pieces, and a marker close to a gene usually travels with it and sometimes does not. So eleven disagreements could be swapping, or a test that misreads. One number cannot separate those two. A second test aimed at the gene itself can.

TAKEAWAY 15W

A test can be right about the marker and still be wrong about the plant.

8.4 What the crew is pulling today

POPULATION GENETICS · GENETIC RESOURCES OFFICE · A PERSON · SPOT

SCENE 28W

The rogueing crew works to one rule, chalked at the end of the row. It gets rewritten when the season turns. Nobody walks the rows to tell them.

GUIDE 43W

Pull what the rule asks for and leave the rest. The chalk changes during the morning and nobody says so. What counts is the plants either side of a change, because they show whether you are reading the board or working from memory.

ASK 15W

Pull to the chalk at the end of the row, and keep checking the chalk.

BEHIND THE BUTTON 66W

Why the rule changes. Early on the crew is pulling anything that is not the line. Once rust arrives they are looking for something else, and the plants that were being pulled an hour ago are now the ones worth keeping.

Why it is hard to spot. A pulled plant leaves a gap and nothing else. Nobody can tell afterwards which rule it was pulled under.

TAKES AS READ

a work rule can be changed while the work is still going on

VERDICT 126W

Three rules run across the morning and each one wants different plants. Anything that is not the line, then anything flowering early, then anything with rust on it. A plant can match two rules at once, and that is what makes the change cost something. What the panel scores is the plants either side of a change. Most of the row is wanted by neither rule, so leaving those alone proves nothing. The damage is hard to see afterwards. A plant pulled under the old rule leaves a gap in the row, and the gap does not say why it is there. That is why the nursery rule is to read the chalk at the start of every row, not at the start of the morning.

TAKEAWAY 18W

The cost of a rule that has changed is paid by whoever is still using the old one.

8.5 Where a tonne of grain comes from

AGRONOMY & PLANT ENERGETICS · DRYING & PROCESSING HALL · A ROOM ·
CALLBACK · BALLPARK

SCENE 32W

The agronomist has eleven tonnes of grain from one field on the weighbridge slip. A trainee has just said the crop got it out of the soil. He would like that settled.

GUIDE 59W

Work out how much of the eleven tonnes is the stuff the plant built out of air. About nine tenths of dry grain is sugar and starch, and all of that came in through the leaves as carbon dioxide. One tile is the share that came out of the soil, which is the small part and not the answer.

ASK 15W

How much of the eleven tonnes came in through the leaves, out of the air?

BEHIND THE BUTTON 113W

What grain is made of. Once it is dry, about nine parts in ten are carbohydrate — sugars and starch, built out of carbon dioxide from the air and water, using light. That is where almost all the weight comes from.

What the soil gives. Minerals, and only a few per cent of the weight. They matter enormously, because a crop short of nitrogen cannot build much of anything, and they are not what the harvest is made of.

Where the water goes. Most of the water a crop takes up leaves again through its leaves as vapour. It is not in the weighbridge figure at all, because the grain is weighed dry.

TAKES AS READ

plants build sugars out of carbon dioxide and water, using light

VERDICT 155W

Dry grain is about nine parts in ten carbohydrate — sugar and starch. The plant builds all of it out of carbon dioxide taken in through its leaves, with water and light. So the part of the harvest that came out of the air is $11 \times 0.9 \approx 9.9$ tonnes, near enough ten. The soil's contribution is the other part: minerals, a few per cent of the weight, a few hundred kilograms across the whole field. That small share is not unimportant — a crop short of nitrogen cannot build much of anything — but it is not what the harvest is made of. The water is not in the figure at all. Most of what the crop took up left again through its leaves as vapour, and the grain on the slip has been weighed dry. So the trainee has it backwards: the field gave the minerals, and the air gave the weight.

TAKEAWAY 16W

A harvest weighs what it weighs because of something that was not there when it started.

A crossing block is a decision about a decade, made in one morning with what is to hand.

What the season established

BEFORE THE DAY — TAG

Catch Nakagawa before the machine starts

Nakagawa, the molecular laboratory lead, has the plate layout for this morning's run. Once the machine starts it is four hours, and the layout is the only thing that says which well holds which line. Without it the result is ninety-six numbers and no names.

PLAN CARD 69W

Friday, and the season is set. The crossing block went in yesterday. WM-712 was released with its weakness written on the label. The drifted batches are on a list that will take fifteen years to work through. The board meets on Monday, and Qureshi has to explain the season to people who fund gene banks. Today you settle what the season established, and what it cost to find out.

BRIEFING 16W

The block is in. The rust is here. The collection is smaller than the catalogue says.

WHAT YOU DECIDE 11W

Say what Wellmere knows now, what it cost, and what changes.

9.1 What the catalogue says and what is there

CONSERVATION & VIABILITY · SEED VAULT · A ROOM · BALLPARK

SCENE 35W

Ninety batches were tested. Thirty had changed, and every one of those had been regrown at least twice. The sixty regrown once or never still match. Nine thousand batches have been regrown twice or more.

GUIDE 53W

Turn the test into a number for the risky group, and only for that group. Every batch that had changed was one that had been regrown twice or more, and the store holds nine thousand of those. One tile is the whole collection, which is the number the board must not be given.

ASK 11W

What number of batches does this season's test put at risk?

BEHIND THE BUTTON 109W

What the test showed. All thirty of the changed batches came from the regrown-twice group, and none of the sixty from outside it had changed. That points hard at regrowing as the risky step.

What it did not show. How many batches nobody tested have changed. Most of the collection has never been regrown at all, and that is the group where nothing was found.

What the honest sentence sounds like. A share, the group it applies to, and the size of that group: about a third of nine thousand regrown batches, which is around three thousand. That is not the same sentence as a third of the collection.

TAKES AS READ

a sample can support a statement about the group it came from

VERDICT 139W

Thirty of the ninety tested batches had changed, and every one of the thirty had been regrown twice or more. So the share that matters is a third, and the group it applies to is the nine thousand batches that have been regrown twice or more: $9,000 \times 0.33 \approx 3,000$. That is the honest number. Doing the same sum on the whole collection gives thirteen thousand, and it is a claim about batches nobody tested — most of which have never been regrown at all, which is exactly the group where nothing was found. Both numbers come from the same test. Only one of them is a statement the test supports, and the board will quote whichever sentence it is given. That is why the group has to travel with the share every time it is written down.

TAKEAWAY 16W

A sample can point at a risky group without proving how many in it are affected.

9.2 One site, one year, again

FIELD TRIALS & SELECTION · FIELD LABORATORY · A PERSON · CHOICE

SCENE 28W

The trial is scored and the lines are chosen. The summary goes to a board that will remember one number. It was a dry season, at one site.

GUIDE 51W

Four ways to write one result. Ask of each where the conditions sit next to the number. A footnote is not the same sentence. The chosen plots are worse as a headline, because they were picked partly for their ground. And holding it back treats a limited result as no result.

ASK 7W

How should this season's result be summarised?

BEHIND THE BUTTON 169W

Why the wrong answers are worth reading. Each one is written to look right to somebody who made a particular mistake. Working out which mistake each one belongs to is most of the learning in this format — more than getting the right one, which you can sometimes do by feel.

Why sounding careful is not the same as being right. A long answer with a lot of detail in it feels safer, and it is easy to write one that is still wrong. The check that builds these questions measures option length across the whole game so the longest one is not the answer more often than chance. Reading is the only way through.

What the verdict adds. After you answer, the verdict says why each wrong option is wrong, in its own words. That is written for the reader who picked it. Read it even when you were right, because being right for a shaky reason is the thing that catches up with you two stops later.

TAKES AS READ

a trial's conclusions belong to the conditions it ran under

VERDICT 68W

Strip the conditions off a yield number and it turns into a claim nobody tested. This season's gain belongs to one site in a dry year. The chosen plots are worse still as a headline. They were picked after the results were seen, so their advantage includes lucky ground. Putting the site and the season in the same sentence is not caution. It is what the number means.

TAKEAWAY 14W

The sentence worth writing is the one that survives being quoted on its own.

9.3 Twelve years, and what they are for

AGRONOMY & PLANT ENERGETICS · DRYING & PROCESSING HALL · A ROOM · BALLPARK

SCENE 32W

Toft's records of cover, flowering and weather now run to twelve seasons. Nobody has ever asked for the whole series. This year two of them explained why the best plots pulled away.

GUIDE 50W

Work out what an ordinary summer's rain is here, by averaging the twelve seasons on the sheet. The total is given; share it out across the number of seasons. One tile is this summer's rain, which is the number you are trying to place rather than part of the average.

ASK 14W

What does an ordinary summer give here, across the twelve seasons on the sheet?

BEHIND THE BUTTON 124W

Why one season cannot be read alone. This summer gave 118 millimetres. On its own that is just a number. Against twelve seasons it turns out to be the second driest, and that is what makes the yield difference mean something.

What an average is for. It says what is ordinary, so anything can be compared with it. Twelve numbers give a fair idea. Two or three would not, because one odd season would pull the answer around.

Why it can only be started early. The weather record can be borrowed from the airfield. Cover and flowering cannot be borrowed from anywhere, because those crops are gone. A measurement nobody asked for in 2014 is the reason this season can be read at all.

TAKES AS READ

a long record lets one season be compared against the others

VERDICT 161W

An average says what is ordinary. The twelve summers on Toft's sheet add up to 2,352 millimetres of rain, so the average is $2,352 \div 12 = 196$ millimetres. This summer gave 118. That is a long way below ordinary — the second driest of the twelve — and it is the fact that makes the rest of the season readable. Without the twelve, 118 is just a number, and the best plots pulling away is just something that happened. With them, the station can say the good plots did well in a season near the dry end of what this place does, which is a sentence somebody can use when choosing what to sow next year. And note which half of the record could have been rebuilt later. The rain can be borrowed from the airfield. Cover and flowering cannot, because those crops are gone, so the measurements nobody asked for in 2014 are the ones doing the work now.

TAKEAWAY 16W

A series of measurements is worth more every year, and it can only be started early.

9.4 An extra generation in the bay

HEREDITY & CROSSING · CROSSING HALL · A ROOM · CALLBACK · SEQUENCE

SCENE 28W

Bengtsson can run one more generation over the winter in the warm bay. A generation there takes three months instead of twelve. There is room for one population.

GUIDE 47W

Three jobs want the bay, and the rail is not a clock. Put them in order by what the winter can never give back: the job that only works indoors goes first, and the job that the field will do just as well in spring goes last.

ASK 16W

Order the three jobs by how much the winter in the bay is worth to each.

BEHIND THE BUTTON 119W

What the bay is good for. Getting a generation finished out of season, so a cross moves one step closer while the fields are empty. Three months instead of twelve, four times a year instead of once.

What the bay is bad for. Judging anything. Plants in a warm bay grow in light, water and temperature nothing like a field, so a yield measured there is a fact about the bay. Any judgement has to wait for ground outside.

Why multiplying seed goes last. It works perfectly well in the bay and it works perfectly well in a field in spring, so spending the winter on it spends a scarce thing on a job that has an easy alternative.

TAKES AS READ

a glasshouse can finish a generation out of season

VERDICT 158W

The bay does one thing nothing else can do. It finishes a generation out of season, in three months instead of twelve. So the job that belongs there is the one where a generation is the whole point. That is the backcross. It needs one more round before it can be tested outside, and every winter it is not run adds a year to the programme. The overdue batch comes next. It does need regrowing, and it needs enough plants to keep what is in it, which the bay can just manage. Multiplying seed for the merchant goes last. Not because it does not matter — it works just as well in a field in spring. Putting it in the bay spends the scarce thing on the job with an easy alternative. And nothing in there can be judged. Yield in a warm bay is a fact about the bay, so any measuring still waits for ground outside.

TAKEAWAY 12W

A generation indoors is for moving material along, not for judging it.

END OF THE DAY 19W

A collection is only as good as the last time each accession was grown, and now Wellmere knows which.

Card text only — no options, boards or answers. 8,925 words of card prose across 40 stops. Generated from themes/seedbank_ms by tools/make-cardbook.mjs.